fn function_report_noisy_top_k

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1 Hoare Triple

Precondition

Compiler-verified

Types consistent with pseudocode.

Caller-verified

`scale` is positive.

Pseudocode

```
def function_report_noisy_top_k(
    k: int, scale: FBig, optimize: str
) -> Callable[[list[TIA]], list[int]]:
    def function(x: list[TIA]) -> int:
        def try_cast(v):
            try:
                return FBig.try_from(v)
            except Exception:
                return None

        # Cast to FBig.  #
        x = ((i, try_cast(x_i)) for i, x_i in enumerate(x))
        # Discard failed casts.  #
        x = ((i, x_i) for i, x_i in x if x_i is not None)

        # Normalize sign.  #
        y = ((i, -x_i if optimize == "min" else x_i) for i, x_i in x)

        # Initialize partial sample.  #
        def partial_sample(shift):
            rv = MO.random_variable(shift, scale)  #
            return PartialSample.new(rv)  #

        y = ((i, partial_sample(y_i)) for i, y_i in y)

        # Reduce to the k pairs with largest samples.  #
        def max_sample(l, r):
            return l if l[1].greater_than(r[1]) else r

        y_top = top(y, k, max_sample) #

        # Discard samples, keep indices.  #
        return [i for i, _ in y_top]

```

```
35     return function
```

Postcondition

Theorem 1.1. Returns a noninteractive function with no side-effects that, when given a vector of non-null scores, returns the indices of the top k z_i, where each $z_i \sim RV(\text{shift} = y_i, \text{scale} = \texttt{scale})$, and each $y_i = -x_i$ if `optimize` is `min`, else $y_i = x_i$.

The returned function will only return an error if entropy is exhausted. If an error is returned, the error is data-dependent.

Proof. Assume the scores are non-null, as required by the returned function precondition. Therefore casts on line 11 should never fail. However, if the input data is not in the input domain, and a score is null, then line 13 will filter out failed casts. This can be seen as a 1-stable transformation of the input data.

The algorithm then proceeds to line 16. Assuming `optimize` is `max`, the line is a no-op, otherwise it negates each score, therefore each $y_i = -x_i$ if `optimize` is `min`, else $y_i = x_i$.

The algorithm proceeds to line 19. The output measure has an associated noise distribution that is encoded into the Rust type system via `SelectionMeasure`. `MO.random_variable` on line 21 creates a random variable `rv` distributed according to `MO::RV` and parameterized by `shift` (the score), and `scale`.

To sample from this random variable, line 22 constructs an instance of `PartialSample`, which represents an infinitely precise sample from the random variable `rv`. We now have an iterator of pairs containing the index and noisy score of each candidate.

The algorithm proceeds to line 26, which defines a reducer based on `PartialSample``greater_than`. Assume the scores are non-null, as required by the returned function precondition. Then `max_sample` on line 26 defines a total ordering on the scores.

Since the score vector is finite, and `max_sample` defines a total ordering, then the preconditions for `top` are met. Therefore on line 30 `top` returns the pairs with the top k scores. Line 32 then discards the scores, returning only the indices, which is the desired output.

If entropy is exhausted, then the algorithm will return an error from `PartialSample``greater_than`. Otherwise, there is one source of error in the function, when there are no non-null scores in the input vector. □

The algorithm avoids the pitfall of materializing an infinitely precise sample in memory by comparing finite arbitrary-precision bounds on the noisy scores until the lower bound of one noisy score is greater than the upper bound of all others.